

## 0.1 Lengthscale and the limits of amplitude transfer

For fixed-smoothness Matérn kernels, equivalent RKHS norms preserve a managed spectral order that already holds; they do not identify its exponent or amplitude from the kernel alone. If  $\mu_j(\ell) \sim c(\ell)j^{-b}$  with common  $b > 1$ , the asymptotic minimizer of the theoretical envelope satisfies

$$\frac{\lambda_T^{\text{bound}}(\ell_2)}{\lambda_T^{\text{bound}}(\ell_1)} = \left( \frac{c(\ell_2)}{c(\ell_1)} \right)^{1/(b+1)} \left( \frac{\rho(\ell_2)}{\rho(\ell_1)} \right)^{b/(b+1)}, \quad \rho = B/A.$$

This is not an oracle-penalty identity. Managed eigenvalues alone do not determine expected-payoff signal or score noise, and two-sided spectral bounds need not supply a limiting amplitude. In the real JKP cache, the initial Matérn finite-range slopes vary across lengthscales and the predeclared amplitude-identification gate fails. Mechanical transfer paths are therefore falsification exercises. The 492 common OOS months for decisions 1984–2024 include fixed lambda, annual CV, joint CV and one-anchor controls; adaptive selection uses only the previous 60 genuinely OOS candidate returns. Actual signed weights reconstruct net costs and paired circular blocks assess differences conditional on fitted paths. The standalone investigation does not establish a convincing structural one-constant contribution for the current paper. We recommend exclusion, while retaining the negative result and its audit. The Gaussian extension tests the same finite-range heuristic while preserving the negative control against an assumed polynomial population law.